Rishikesh Ombase

Agentic AI Architect · Senior Data Scientist

I design multi-agent AI systems with MCP tool layers, RAG and memory, backed by the evals, guardrails and observability that make them safe in production.

rishikeshombase4@gmail.com
in/rishikesh-ombase-11989b1ab
github.com/Rishikesh32
rishikesh32.github.io
Kolhapur, Maharashtra, India

Profile

Agentic AI architect and Senior Data Scientist with 6+ years building machine learning and LLM systems, with deep experience in **financial and trading data**. I design agentic systems end to end: multi-agent orchestration with LangGraph, MCP tool layers, agentic RAG and long-term memory, and model routing across frontier and open-weight models. Every design is backed by **evals, guardrails and observability**, with humans approving anything irreversible.

Experience

Senior Data Scientist
Dec 2023 – Present
AIT Global India

- Designed and built LLM-integrated API services and automated pipelines that extract insights from financial reports using Python, OCR and SQL.
- Developed generative AI modules for financial commentary summarisation and sentiment analysis.
- Collaborated on risk-scoring models with AI-augmented forecasting capabilities.
- Designed and deployed the APIs behind portfolio intelligence dashboards.

AI Data Specialist
Jul 2023 – Oct 2023
Futurense Technologies

- Built real-time web data collection that enriched datasets for pricing models and market-trend analysis.
- Delivered executive decision dashboards with Python, Power BI and SQL.

Software Engineer
Jul 2022 – Mar 2023
Systech Solutions

- Designed data frameworks for large-scale trading datasets on AWS.
- Enabled real-time updates to the analytics dashboards that monitor asset performance.

Data Scientist
Jan 2020 – Mar 2022
Varilligence Technologies

- Researched and built machine learning models for stock-price movement prediction using PySpark and financial indicators.
- Built alpha-generating trading signals from historical equities data, macro indicators and sentiment variables.
- Contributed to model validation and A/B testing of quant strategies in simulation environments.

Highlights

6+ years in AI, ML and data

4 companies, finance-heavy

7 agentic architectures

M.Sc. Computer Science

Expertise

Agentic AI

LangGraph · MCP servers · Multi-agent systems · CrewAI · OpenAI Agents SDK · Claude Agent SDK · A2A · Computer use · Human-in-the-loop

Knowledge & memory

RAG · GraphRAG · Hybrid search · Reranking · LlamaIndex · Qdrant · pgvector · Neo4j · Agent memory · Context engineering

Models & ML

Claude · GPT · Gemini · Llama · Qwen · DeepSeek · Vision-language models · LoRA / QLoRA / DPO · Model routing · PyTorch · scikit-learn · Forecasting · OpenCV

LLMOps & governance

Evals · Ragas · DeepEval · Langfuse · LangSmith · Guardrails · Prompt-injection defence · OpenTelemetry

Platform & cloud

Python · FastAPI · Pydantic · Docker · Kubernetes · vLLM · Ollama · Redis · PostgreSQL · AWS Bedrock · Azure AI Foundry · Vertex AI

Selected projects

AlphaDesk

Multi-agent quant research platform

In progress

- Hierarchical LangGraph supervisor that delegates to market-data (MCP servers), research (GraphRAG over filings and news), quant (sandboxed forecasting and backtests) and risk agents.
- Model routing between frontier and open-weight models; every run traced in Langfuse and gated by an eval suite.
- Human approval required before any signal leaves the system.

Agent flow

Supervisor → Market data · Research · Quant · Risk → Report → Human approval

LangGraph · MCP · GraphRAG · Qdrant · Neo4j · Code sandbox · Langfuse · Kubernetes

FundDocs Studio

Agentic styling for fund filings

Document AI

- Multi-agent pipeline that takes prospectuses, summary prospectuses and SAI documents with the client's style guide.
- Vision-language models read each page's layout; a style-guide agent decides the rule for every heading, table and footnote; LLM agents apply and verify formatting, then route for human sign-off.

Vision-language models · LLM agents · LangGraph · Style-guide RAG

ListingStudio

Multimodal real-estate listing generator

Multimodal

- Classified property images by room and feature, then generated listing content at scale with LLMs: views and surroundings, amenities, neighbourhood details and full descriptions.
- Critic step checks generated claims against photos and listing data before publishing.

Vision LLM · GPT models · FastAPI · Structured outputs

CineMind

OTT recommendation engine

Recommendations

- Grouped viewers into taste clusters from watch history so similar users receive similar films; cluster and item-similarity candidates ranked per user.
- Redis serves precomputed recommendations, sessions and trending lists; FastAPI microservices on PostgreSQL.

FastAPI · PostgreSQL · Redis · Embeddings · Clustering · Docker

Invoice Data Automation

Document AI workflow

NLP · OCR

- Automated extraction and cleaning of invoice data from Word and PDF documents with NLP and OCR.
- Filled target systems automatically, improving processing speed and accuracy.

Python · OCR · NLP · Selenium · Scheduling

Earlier work: drone-type recognition with computer vision on Apache Spark · FOX Channel data integration for the Looker BI team · automated mutual-fund NAV pipeline · quantitative stock-price forecasting.

Education

M.Sc. Computer Science
Shivaji University, Kolhapur · 2022

B.Sc. Computer Science
Mahavir College, Kolhapur · 2019

Recognition

- Best Performer of the Month
- Spot Award
- GOM Award

Certifications

- Machine Learning in Python
- Data Analysis with Python
- Data Visualization